

REPORT

Tijuco Preto WWTP

Appendix 4.2: Control Narrative

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Version History

Version	Date	Key Changes
1	16 June 2017	First issue to the client.
2	17 July 2026	Updated to the latest standard

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1 INTRODUCTION

1.1 General

This document describes the process control of the Nereda® reactors of the Tijuco Preto WWTP (Waste Water Treatment Plant). It is based on the P&ID's and the document "Control Philosophy", belonging to this project.

This document is a base document for the technical design of the software of the PLC control for the Nereda® process.

It is noted that based on the functionality described in this document, the customer has to prepare a technical design that is eventually based on its own process automation standards.

1.2 NEREDA® Controller

Control of the Tijuco Preto Nereda® WWTP is realized with a PLC in combination with the Nereda® Controller. This document describes the functionality of the PLC software.

In the document "Control Philosophy" (appendix 4.1), the interaction and interfaces between Nereda® Controller and PLC are detailed. This document has to be read first before the document is processed into technical design. The software for the Nereda® Controller itself is supplied by RHDHV.

The hierarchical structure is depicted in the figure below:

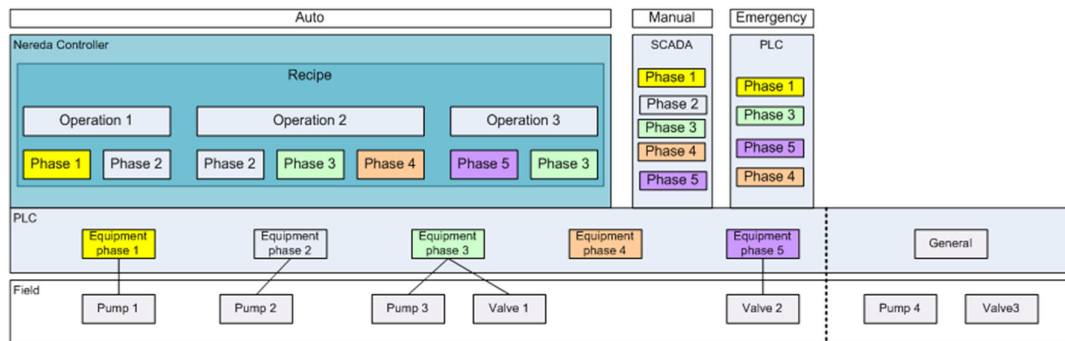


Figure 1 – Structure of Nereda® Controller, process automation and field

A complete list of data to be exchanged between Nereda® Controller and PLC will be made available in Excel format.

1.3 Process parts

The installation consists of the following process components:

- Influent feed
- Nereda® reactors
- Aeration
- Sludge buffer

1.4 General principles

Here some standard control principles are described which may be considered as a guideline for the elaboration of the technical design. These are commonly used basic functions in PLC-software applications. Slight deviations from this are possible, depending on the customer process automation standards.

1.4.1 Valves

The control of valves is incorporated in various sequences. For valves, the following items should be presented:

- position open / closed
- opening / closing (neither of limit switches active)
- failure

Opening and closing of valves should be monitored (principle: ACOF: Automatic Check Of Feedback, safeguarding). The Open and Closed position readings should follow the Open and Close commands and the combination of the Opened and Closed readings should be valid. If a valve does not reach the position in which it is controlled, an alarm should be produced that gives an accurate description of the situation. The default action after an alarm occurs is Close (in case of 'fail safe' position).

1.4.2 Ready status

An equipment has a status 'ready' or 'available' when it is switched via SCADA in Auto mode and it has no failures or alarms.

1.4.3 Phased restart after power failure

The control system should be equipped with a phased (re) start mechanism (after power failure). This implies that devices a certain time after each other are put into operation. Upon coming into operation of a device, a hold signal is activated for several seconds. As long as the hold signal is present, no other device can be started.

After power return the control system should wait an adjustable time (0- 5 minutes) before it generates a general reset command (initializing time for equipment). Then the equipment can be put into operation via phased restart.

1.4.4 Feedback monitoring

For each automatically or manually given command by the controller (e.g. on/off, start/stop), it has to be checked whether this operating command is executed. The On / Off or Start / Stop readings should follow the commands, the combination should be valid. If not, an alarm should be produced that gives an accurate description of the situation. The default action after an alarm occurs is Off or Stop.

Feedback monitoring is not listed separately on the specification sheets.

1.4.5 Measurement failure

Measurement failure is derived from:

- the digital input 'malfunction measurement'
- monitoring of the 4-20 mA signal $<3.5 \text{ mA}$ and $> 21 \text{ mA}$
- or a combination of the above.

Each failure of a measurement must be reported separately.

1.4.6 Operation level hierarchy

On the SCADA, control is possible via a software switch Manual-0-Automatic the 'Automatic' control is determined by the automation of the PLC control system.

The control of the plant has three operating levels, automatic, manual and field:

1. Automatic: via SCADA and PLC. The automation system controls the equipment on base of measurements in the field and supervises the process.
2. Manual: via SCADA the operator can intervene in the automatic control. Starting and stopping of equipment is possible. The automation system is still supervising the process. Safety and machine interlocking signals will still be active.
3. Field: by means of a local switch, equipment can be started and stopped unconditionally.

The priority is: field control, manual, automatic, with field control having the highest priority and automatic the lowest priority. On the SCADA, selection of manual or automatic operation is possible via a software switch Manual-0-Automatic

1.4.7 Alarm handling

Any failure must be processed in an alarm. Alarm handling has to take place by the acceptance and acknowledgment of alarms. When an alarm is accepted it may only be reset if the cause of the alarm is no longer present.

1.4.8 Settings

Minimum flow rates, waiting times and so on, as specified in the control narrative must be adjustable via SCADA. For parameters that need to be adjustable only via the PLC, the text '[PLC]' is added.

Signals from pressure switches, level measurements (both discrete inputs and software derived values), flow switches, etc., which have the effect of an alarm, should have a standard adjustable input delay of 0-15 seconds.

1.4.9 Alternating and standby operation

Where pumps are specified for duty + standby operation, the control part must be implemented as follows.

Alternating operation

When a pump must be put in operation, the pump that has been out of operation for the longest time, is selected as duty pump. If several pumps are running and one pump should be out of operation, the pump must be selected that is already in operation for the longest time.

The control system will also rotate the duty pumps by monitoring the run hours of pumps. Pumps which have been taken out of order for any reason will not be selected by the PLC / control system. Selection of the duty and standby pumps will normally be a PLC function (AUTORUN) by means of duty rotation program.

Standby operation

For pumps or other equipment that have standby operation, on SCADA a choice has to be made between pump 1, 2 or automatic alternating operation (duty rotation program in PLC).

In case of failure with pumps with duty+standby configuration, an alarm is generated and the standby pump will be put in operation. Also, if the preset selected pump is not available (anymore), the other pump will take over the function.

1.4.10 Cyclic 'on/off' or 'pulse/pause' operation

Based on this cyclic control principle, a device is put into operation during the specified 'on' (pulse) time and taken out of operation during the 'off' (pause) time. For each device, both times must be adjustable and set separately via the SCADA system.

A pulse/pause or on/off cycle should always begin with the 'on' (pulse) time. Once a device is put into 'AUTO' mode from the SCADA-system or a cabinet switch, the cycle must start again.

If a value "0" is filled in for the 'off' (pause) time, this should be processed as continuously on. If a value "0" is entered for the 'on' (pulse) time, this should be processed as continuously off.

1.5 Explanation specification sheets

In the description of the installation, the process control is divided into a number of process components. The following items are listed for each process element:

- automatic control that include multiple equipment associated with the process part
- equipment associated with the process part
- instrumentation associated with the process part

Each part of the process is described in a separate chapter. Each chapter begins with a description of the process component. Then, the equipment and instrumentation associated with the process part are described on specification sheets.

The layout of the description of a process part and of the specification sheets is explained below.

It is noted that any customer process automation standards may complement the basic specification sheets included in this document.

1.5.1 Description of a process part

Description

General description of the operation of the process part

Automation

Functional description of the control of the process part, as a single part and in relation to other process parts. These include sequences and controls that affect multiple machines and suchlike.

Settings

Operating options in SCADA for this process part (group commands).

Interlocks

Descriptions of process interlocks, i.e. situations and signals that affect the normal operation. The signals mentioned here switch off the process part, lock it until the relevant signal is dropped and the process part is reset, and generate an alarm. If necessary, differences in manual mode and automatic mode will be indicated.

Presentation

Description of signals and / or events, to be presented for this process part, provided they do not belong to a measurement or machine.

1.5.2 Equipment specification sheet

Settings

SCADA

Operating options in SCADA for this equipment. Also any controller settings are listed here.

Automation

The normal operation of the equipment. In automatic mode, the machine must be switched on or off based on the conditions specified in the specification sheet. This also includes duty cycle / standby algorithms and suchlike.

In Manual mode the machine will operate continuously. Exceptions to this rule are described here.

Locking means that operation of equipment is blocked until the locking conditions are no longer present. After this, the equipment can come back in operation.

Interlocks

Description of the situations and signals that interfere with the Manual and Auto operation mode in normal operation. The signals mentioned here switch off the equipment, lock it until the relevant signal is dropped and locking is reset, and generate an alarm. If necessary, differences in manual mode and automatic mode will be indicated.

Presentation

Description of the presentation of operating status, switch settings and suchlike.

1.5.3 Instrumentation specification sheet

Presentation

Signals to be presented locally or centrally

Switching and alarm points

The following items are presented in table form:

- switching points :
the various switching points of a measurement. These can be hardware contacts or software derived contacts from the measurement signal.
- action :
the actions to be executed when a switching point is triggered
- adjustable (Ad):
indicating that the switching or alarm point has to be adjustable on the SCADA
- preference value (Pv):
default value of the switching or alarm point.
- alarm (Al):
indication that the switch has an alarm function.
- alarm text :
alarm text that has to be mentioned in the alarm list.

A specification table looks like this:

SCADA

- flow [m³/h]

switch points	action	Ad	Pv	Al	alarm text
LL	Feed pump off	x		x	Flow too low (time delayed)
L		x		x	Flow low
Failure measurement				x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2 INFLUENT FEED

2.1 Introduction

The influent buffer provides the supply to the Nereda® reactor during the feeding of the Nereda® reactor. In the influent buffer 3 frequency controlled pumps are installed, the pump configuration is 2 duty + 1 standby.

The desired flow rate is determined by the Nereda® Controller. Within the Nereda® Controller is a prediction algorithm is used to predict the influent flow and to keep the optimal level in the influent buffer.

The Nereda® controller determines the flow rate set-point based on the level in the influent buffer and passes this to the PLC.

The influent in the influent buffer tank is continuously mixed with a mixer. Mixing is stopped if the level in the influent buffer is lower than an adjustable level.

2.2 Automation

Reactor feed flowcontrol

The (duty) pump must be started when the 'feed' phase of the Nereda® reactor is in RUN. To control the influent feed flow to the reactor, the feed pumps must be PI-controlled on the flow setpoint from the Nereda® Controller.

If one pump is in operation at an adjustable maximum capacity, the second pump must be started. The capacity of the first pump is lowered to an adjustable capacity C2 then the second pump is started with capacity C2. After acceleration both pumps are controlled together on the flow setpoint. When two pumps are running at a capacity below an adjustable value C3, one pump must be stopped.

Selection of the duty and standby pumps will be by means of a duty rotation program in the PLC.

Flow measurement failure

In case of flow meter failure, corrective actions must be taken in the PLC. The Nereda® Controller will still issue a flow setpoint to the PLC. In the PLC the flow setpoint must be translated into a pump speed by using the theoretical pump capacity.

Interlocking

The flow control of the influent feed must be started when the equipment phase "Feed" of the Nereda® reactor is in the state RUN. When the feed phase is no longer active, the pump(s) must be stopped.

The feed to the reactor must be stopped in the following situations:

- Low Low level influent buffer;
- pH value influent buffer too low or too high;
- conductivity influent buffer too low or too high.

If the measured values for pH and conductivity are back inside the desired range, then the feed must be started again after an adjustable waiting time.

2.2.1 Settings

- settings for flow control
- settings for on/off switching of pumps and mixer
- settings for pH influent buffer too High / too Low
- settings for conductivity influent buffer too High / too Low

2.2.2 Interlocking

- None

2.2.3 Presentation

- actual wait time restart feed pumps

2.3 Specification sheets

2.3.1 Influent feed pump P4001, P4002, P4003

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- Capacity setpoint (Manual)
- Controller settings (gain, integration time)
- settings Low level for dry run protection

Automation

Automatic

In automatic mode, the (duty) pump is controlled as described in section 0.

Manual

In manual mode, the operator can start and stop the pump. The pump must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- thermal overload motor
- moisture detection pump (MS) (no machine stop)
- electric circuit failure
- safety switch off
- frequency inverter failure

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure
- Actual pump capacity

2.3.2 Mixer R4005

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- settings Low level for dry run protection

Automation

Automatic

In automatic mode, the mixer is operated continuously. The mixer is stopped if the actual influent buffer level is lower than an adjustable Low level.

Manual

In manual mode, the operator can start and stop the mixer.

Interlocks

- thermal overload motor
- electric circuit failure
- safety switch off

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure

2.3.3 Level measurement influent buffer 40LT0001

SCADA

- Level [m]

switch points	action	Ad	Pv	Al	alarm text
Start emergency recipe	Start feed phase at Emergency recipe mode	x			
H				x	Level too High influent buffer
LL	Stop influent feed pump + mixer	x		x	Level too Low influent buffer
Failure measurement				x	Failure level measurement influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2.3.4 Level switch influent buffer 40LS0002

SCADA

- Level switch point

switch points	action	Ad	Pv	Al	alarm text
HH				x	Level too High influent buffer
Failure measurement				x	Failure level switch influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2.3.5 Flow measurement reactor feed 40FT0003

SCADA

- Flow [m³/h]

switch points	Action	Ad	Pv	Al	alarm text
LL	Stop influent feed	x		x	Flow too low (time delayed)
L		x		x	Flow low
Failure measurement				x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2.3.6 Conductivity measurement influent buffer 40QT0004

SCADA

- Conductivity [mS]

switch points	action	Ad	Pv	Al	alarm text
L	Stop influent feed	x		x	Conductivity too Low influent buffer
H	Stop influent feed	x		x	Conductivity too high influent buffer
Failure measurement				x	Failure conductivity measurement influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2.3.7 pH measurement influent buffer 40QT0005

SCADA

- pH [-]

switch points	action	Ad	Pv	Al	alarm text
L	Stop influent feed	x		x	pH too Low influent buffer
H	Stop influent feed	x		x	pH too High influent buffer
Failure measurement				x	Failure pH measurement influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3 NEREDA® REACTOR

3.1 Introduction

From the influent buffer the wastewater is pumped batch wise to the Nereda® reactors. The operation of the Nereda® reactors is controlled by a PLC and the Nereda® Controller.

If the communication between the Nereda® controller and the PLC should fail, it switches to an adjustable Fallback Recipe that is programmed into the PLC. See Chapter **Error! Reference source not found.**7 and the description in the document “Control Philosophy” (appendix S 4.1).

3.2 Automation

The processes in the Nereda® reactors and Sludge buffers are controlled by the Nereda® Controller. This control structure is modular. There are modules for the following processes:

3.2.1 Phase feed

In this phase, the reactor is filled via the influent feed pump(s). The switching and control of the pump(s) has been described under the process section “Influent feed” and should not be programmed as part of the feed phase, but as external control logic which is triggered if a feed phase is in RUN. The current paragraph is only the description for the starting and stopping of the feed phase. The feed phase controls reactor feed valve AV41101.

Sufficient feed flow

During the RUN state the level in the reactor must be monitored to ensure a raising level due to the incoming flow. This must be done by comparing the level value at the start of the RUN state (L_{start}) and the Level value after an adjustable amount of time (L_T). If the level difference $L_T - L_{start}$ is less than an adjustable value, the feed phase must be aborted. In the operating conditions this is referred to as “sufficient feed flow”.

Parameters

- Flow ‘feed’ setpoint [m^3/h]

Start logic

- Open feed valve AV41101

RUN logic

- The pump will be operated on the flow setpoint parameter from the Nereda® Controller

Stop logic

- Close feed valve AV41101

Operating conditions

1. Feed valve AV41101 is available
2. Sludge discharge valve AV41103 is available OR closed
3. Lower level valve AV41102 is available OR closed

4. *Reserved, do not use*
5. No HH-level (feed) in the reactor 41LT1005
6. Level measurement 41LT1005 available
7. *Reserved, do not use*
8. Sufficient feed flow (definition: see above)
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.2 Phase lower level

The lower level phase controls lower level valve AV41102. The water from the level correction will be discharged back to the influent buffer.

Parameters

- None

Start logic

- Open lower level valve AV41102.

RUN logic

- None

Stop logic

- Close lower level valve AV41102.

Operating conditions

1. Lower level valve AV41102 is available
2. *Reserved, do not use*
3. Sludge discharge valve AV41103 is closed.
4. No High level in the influent buffer 40LT0001
5. No HH-level in the influent buffer 40LS0002
6. *Reserved, do not use*
7. *Reserved, do not use*

- 8. Spare
- 9. Spare
- 10. Spare
- 11. Spare
- 12. Spare
- 13. Spare
- 14. Spare
- 15. Spare
- 16. Spare

3.2.3 Phase aerate

During the aeration phase, air is brought into the reactor. The desired capacity (0-100%) is specified by means of a parameter from the Nereda® Controller. In the PLC this capacity must be translated into activation and control of the aeration system. The aeration system consists of a combination of blowers and valves.

The selection, switching and control of the blowers and valves, is described in the “blower section” in chapter 4 and should not be programmed as part of the phase, but as external control logic which is triggered if an Aeration Phase of a reactor is in RUN.

Airlock sludge decant

By means of aeration, the sludge discharge internal can be filled with air to create an airlock. This is indicated by a separate parameter from the Nereda® Controller. The air fill valve must be opened when this parameter from the Nereda® Controller is active [=1] and closed when the parameter is not active [=0]. If the parameter is active and the valve has not opened within an adjustable time, an alarm should be generated. The phase should remain in ‘RUN’.

Parameters

- Aeration capacity setpoint [0-100%].
- Air fill valve sludge discharge AV41171 open [0/1].

Start logic

- None

RUN logic

- Control blower(s) with capacity setpoint from the Nereda® Controller.
- During RUN, the valve AV41171 is opened or closed, when the status of the parameter is changed.

Stop logic

- Close valve AV41171

Operating conditions

1. *Reserved, do not use*
2. *Reserved, do not use*
3. Vent valve aeration grid AV41107 is closed
4. *Reserved, do not use*
5. *Reserved, do not use*
6. Air fill valve sludge discharge AV41171 is closed OR (air vent valve AV41172 AND sludge discharge valve AV41103 are closed)
7. Lower level valve AV41102 is closed.
8. NOT (High level (aeration) in reactor 41LT1005) OR phase 'aerate' ('starting' or 'RUN' or 'stopping'))
9. Level measurement 41LT1005 available OR phase 'aerate' ('starting' or 'RUN' or 'stopping'))
10. *Reserved, do not use*
11. *Reserved, do not use*
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.4 Phase metal dosing

The metal dosing phase is used to dose metal salts into the reactor.

Parameters

- Metal salt dosing flow setpoint [L/h]

Start logic

- Open chemicals valve AV41108

RUN logic

- Metal salts dosing flow must be controlled on the flow setpoint parameter from the Nereda® Controller

Stop logic

- Close chemicals valve AV41108

Operating conditions

1. Metal salts valve AV41108 is available and valves for other reactors (AV42108) are closed.
2. Metal salt dosing not active for another reactor.
3. No error from the metal salt dosing unit.
4. Spare
5. Spare



6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.5 Phase wait

Parameters

- None

Start logic

- None

RUN logic

- None

Stop logic

- None

Operating conditions

1. Spare
2. Spare
3. Spare
4. Spare
5. Spare
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare

13. Spare

14. Spare

15. Spare

16. Spare

3.2.6 Phase sludge discharge

The sludge discharge phase controls discharge valve AV41103.

Parameters

- None

Start logic

- Open discharge valve AV41103

RUN logic

- None

Stop logic

- Close discharge valve AV41103

Operating conditions

1. Discharge valve AV41103 is available and valves for other reactors of the same line (AV4x103) is closed
2. Lower level valve AV41102 is closed.
3. *Reserved, do not use*
4. Phase 'feed' sludge buffer (1 for line A, 2 for line B) NOT aborted and NOT interlocked (1 for line A, 2 for line B)
5. Air fill valve sludge discharge AV41171 is closed
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.7 Phase vent aeration grid

This vent phase controls the venting of the air grid with valve AV41107.

Partial pressure relief aeration system

Depending on the type of aeration elements, a certain back pressure may be required. This can be achieved by closing the aeration valve AV41107 during the 'RUN' of the vent phase as soon as sufficient pressure reduction has been established (by means of 41PT1014).

Parameters

- None

Start logic

- Open vent valve AV41107

RUN logic

- If needed by aeration plates: close vent valve AV41107 if pressure gets below a set minimum pressure

Stop logic

- Close vent valve AV41107

Operating conditions

1. Reserved, do not use.
2. Vent valve AV41107 is available
3. Reserved, do not use.
4. Reserved, do not use.
5. Reserved, do not use.
6. Reserved, do not use.
7. Reserved, do not use.
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.8 Phase vent sludge grid

This vent phase controls the venting of the sludge grid with valve AV41172.

Parameters

- None

Start logic

- Open vent valve AV41172

RUN logic

- None

Stop logic

- Close vent valve AV41172

Operating conditions

1. *Reserved, do not use.*
2. Vent valve AV41172 is available
3. *Reserved, do not use.*
4. *Reserved, do not use.*
5. Air supply valve AV41171 sludge grid is closed
6. *Reserved, do not use.*
7. *Reserved, do not use.*
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

3.2.9 Settings

Adjustable values, as mentioned in the automation section above, must be made adjustable on SCADA.

3.2.10 Alarms

The status "Aborted" of each phase must be configured as an alarm

3.2.11 Presentation

Presentation via the web pages of the Nereda® Controller and as indicated in the document "Control Philosophy" (appendix 4.1).

3.3 Specification sheets

3.3.1 Valves reactor AV41101-41103, AV41107-41108, AV41171-41172

Settings

SCADA

- Manual-0-Auto
- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under chapter 3.2.

Manual

In manual control the operator can issue the following commands:

- open
- close

Interlocks

- check on feedback failure (valve opening / closing)
- electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

Failure of feed valve AV41101

If the isolation valves HV..151 and HV..154 of valve AV41101 are electrically operated (optional, to be determined by the customer), one of these valves can be used as a replacement feed valve if the AV41101 fails.

Failure of sludge discharge valve AV41103 or WLC valve AV41102

If the isolation valves HV..176 and HV..177 of the valves AV41103 and AV41102 are electrically operated (optional, to be determined by the customer), the corresponding electrically operated insulation valve can be used as a replacement valve if AV41103 or AV41102 fails.

3.3.2 Dry solids measurement reactor 41QT1001

SCADA

- Dry solids [g/l]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement				x	Failure dry solids measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.3 Temperature measurement reactor 41QT1002

SCADA

- Temperature [°C]

switch points	Action	Ad	Pv	Al	alarm text
H	-	x		x	Temperature reactor high
Failure measurement				x	Failure temperature measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.4 Oxygen measurement reactor 41QT1003

SCADA

- Oxygen concentration [mg/l]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure oxygen measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.5 Redox measurement reactor 41QT1004

SCADA

- Redox potential [mV]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure redox measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.6 Level measurement reactor 41LT1005

SCADA

- Level[m]

switch points	Action	Ad	Pv	Al	alarm text
HH	Abort / interlock Feed phase -> operation conditions not met	x		X	Level too High to feed
H	Interlock Aerate phase reactor-> operation conditions not met	x		X	Level too High for aeration
FB_RL1 Start aeration	Stop Lower Level phase reactor during Fallback mode, see FB-chapter	x			
FB_RL2 Sludge discharge	Stop Sludge discharge phase reactor during Fallback mode, see FB-chapter	x			
Level difference sludge discharge	Monitoring the amount of discharged sludge during phase 'sludge' discharge. See Error! Reference source not found.	x		x	Insufficient sludge discharge
LL	High priority alarm (reactor level too low, possible leakage)	x		x	Reactor level too low," leakage"
Failure measurement	Abort / interlock Feed, Aerate phase -> operation conditions not met			x	Failure level measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.7 Ammonia NH₄ measurement reactor 41QT1007

SCADA

- Ammonia concentration [mg/l]

switch points	Action	Ad	Pv	Al	alarm text
H	High ammonium alarm (only enabled during 'Feed' phase)	x		x	NH ₄ breakthrough during 'Feed' phase.
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure ammonia measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.8 Nitrate NO₃ measurement reactor 41QT1008

SCADA

- Nitrate concentration [mg/l]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure nitrate measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.9 Phosphate PO₄ measurement reactor 41QT1009

SCADA

- Phosphate concentration [mg/l]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure phosphate measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

4 Blower Section

4.1 Introduction

The blower section controls the blowers and valves to provide the reactor with sufficient air. For the aeration of the Nereda® reactors, 4 frequency controlled blowers are installed.

The blower configuration is 3 duty + 1 standby, , therefore the configuration is limited to a maximum of 3 blowers in simultaneous operation.

The number of blowers in operation and the capacity of these blowers is controlled based on the air demand of a reactor, determined as described below.

Note that the Process Automation-system must ensure that the blowers and valves are switched correctly during states 'starting' and 'RUN' of the aeration phase. It is possible that 2 aerated phases can occur one after the other in a cycle, the statuses must then be able to switch from RUN-STOPPING-READY to STARTING-RUN again without any problems.

4.2 Automation

The blower system includes the following components:

- the selection valves AV46001, AV46002 and AV46003
- blowers V4601, V4602, V4603 and V4604

The aeration phase of each reactor provides a START command, STOP command and a Capacity setpoint for the blower section.

*Please note that this is **not** the same type of control as a frequently applied system based on oxygen demand (feedback control principle). The Nereda system determines the air demand based on different control modes and various parameters such as the NH_4 , NO_3 , Redox and O_2 -measurements in the reactors.*

Each reactor will demand a maximum of two blowers. The number of requested blowers (1 or 2) is determined by means of the capacity setpoint from the Nereda® Controller. The setpoint from the Nereda® Controller is in the range 0..100%. The blower capacity has to be scaled to the requested aeration capacity from the Nereda controller. On SCADA these settings have to be adjustable.

- 0% demand from the Nereda® Controller means 0 Nm³/h.
- 100% demand from the Nereda® Controller means two blowers at maximum air demand reactor.

Following table shows an example for the aerating capacity at peak air demand during DWF conditions.

Table 1 – Example scaling air capacity - blowercapacity

Blower capacity	Air capacity setpoint Nereda® Controller	Air flow reactor
1 blower 0 – 100 %	0 – 50%	4.190 Nm ³ /h (max air capacity 1 blower)
2 blowers 0 – 100%	50% – 100%	8.370 Nm ³ /h (max. air demand reactor)

Calculation help. 0-50% => 100*(4190/8370)

0-100% => 100*(8370/2*4190)

A reactor that sends a start command must obtain at least one blower.

Below is a guideline for selecting the correct valves and blowers. The steps must be followed continuously in the Process Automation software to determine the optimal blower and valve configuration.

Step 1

Determine number of blowers, based on Section **Error! Reference source not found..**

Step 2

Skip, not applicable for this header configuration with 1 header per reactor.

Step 3

In this step the possible blower configurations must be determined, based on the position of the selection valves. The possible blower configurations are listed in **Error! Reference source not found..** In the last 2 columns the possible blowers for that header are indicated. (note that these are the blowers that can be selected for operation, the actual number of blowers to be put into operation is always limited by the number of duty blowers that may be in operation simultaneously).

Table 2 - Possible blower configurations

Configuration	Blower selection valves			Header 1	Header 2
Nr.	AV46001	AV46002	AV46003	Blowers	Blowers
1	close	close	close	V4601	V4604
2	close	close	open	V4601	V4603 V4604
3	close	open	close	V4601	V4604
4	close	open	open	V4601	V4602 V4603 V4604
5	open	close	close	V4601 V4602	V4604
6	open	close	open	V4601 V4602	V4603 V4604
7	open	open	close	V4601 V4602 V4603	V4604
8	open	open	open	V4601 V4602 V4603 V4604 (not allowed)	V4601 V4602 V4603 V4604 (not allowed)

Step 4

From the possible blower configurations exclude those configurations that are not possible. This results in a table with allowed configurations.

- Configurations must be excluded when **Error! Reference source not found.** indicates which configurations must be excluded when a valve is in alarm.
- The manual position of a valve does not match the required position of the valve (e.g. *hand / open*).

Table 3 – Excluded configurations

AV46001 alarm	AV46002 alarm	AV46003 alarm	Configuration number to exclude (Error! Reference source not found.)
0	0	0	-
0	0	1	2 4 6 8
0	1	0	4 7 8
0	1	1	2 4 6 7 8
1	0	0	5 6 7 8
1	0	1	2 4 5 6 7 8
1	1	0	4 5 6 7 8
1	1	1	2 4 5 6 7 8

If a valve alarm is reset and the valve is manually switched into the open or closed position, the corresponding configuration is allowed again (e.g. AV46003 alarm will exclude configurations 2, 4, 6 and 8).

Step 5

For each allowed blower configuration calculate the number of blowers for each header. Only count blowers with the status 'available'. An example of a table is shown below.

Table 4 – Example table for header capacity (with Blower V4601 in error)

Configuration	Header 1	Header 2
No.	Blowers	Blowers
1	-	1
2	-	2
3	-	1
4	-	2 (1 not used)
5	1	1
6	1	2
7	2	1
8	-	-

Step 6

Determine if the current configuration is insufficient.

The current configuration is insufficient when:

- the current configuration is excluded
- the capacity of the current configuration no longer fits the need of the aeration phase(s) and by switching reactors and headers it does

When the current configuration is insufficient continue with Step 7, else start again at Step 1.

Step 7

The requested number of blowers from Step 1 must now be combined with the allowed blower configurations from Step 5.

Select the allowed configuration depending on the number of requests:

- one request: select the blower configuration with the highest number (2) of blowers available
- two request: the reactor that demands the highest capacity must be assigned to the blower configuration with the highest number of available blowers

The preferred configuration in this example is configuration 6.

If a reactor requests a blower and there is no configuration possible to switch a blower on this reactor, the signal 'insufficient capacity' for this reactor must be set (*Note: this signal must be suppressed during stopping of the aeration phase*).

If the air capacity of a reactor is lower than requested for an adjustable time, an alarm message 'insufficient air capacity' must be generated (*Note: this signal must be suppressed during stopping of the aeration phase*).

Continue with Step 8 if the new configuration is different from the current configuration. If no new configuration is possible, maintain the current configuration and start again at Step 1.

Step 8

Switch the selection valves in the corresponding position and start the blowers on the requested capacity.

When executing a switch between configurations it must be prevented that blowers run against a closed valve. This must be done as follows:

- switch running blowers to minimum rpm
- open all selection valves
- stop blowers that should not be in operation in the new configuration
- switch selection valves to the position for the new configuration.

(The blowers will be provided with a (mechanical) pressure relief valve, but the use of that valve must be limited to emergency events.)

If a stop command is received from a reactor the corresponding blowers must be stopped.

4.2.1 Blower control

The blowers are started when the Aeration Phase of a reactor is in RUN. The number of blowers in operation on the header will be primary determined by means of the requested air capacity by the Aeration phase in STARTING/RUN, see section **Error! Reference source not found.** In this way extra delay due to switching on/speeding up blowers one by one after each other, is avoided.

As soon as the aeration of a reactor starts, the total air demand will increase by a certain percentage. Based on this total air demand and the blower capacity ranges it must be determined how many blowers should be in operation to be able to meet the desired total air demand.

A typical procedure for switching blowers on/off to the header system is described below. The procedure may be changed in respect to the blower's technical specifications.

Blower switching on

IF

- The requested total air capacity is higher than the total maximum capacity of the blowers in operation
- THEN the blower(s) must be switched into operation as follows:
- Ramp down running blower(s) to an adjustable capacity C2
- Start the (additional) blower(s) with capacity C2
- When the blowers are in operation for an adjustable time, control them on the requested air capacity (working as a single unit).

Blower switching off

IF

- Multiple blowers are in operation
- AND the requested air capacity is lower than the total minimum capacity of the blowers in operation
- AND the blowers are running at a capacity below an adjustable value C3 for an adjustable time,

THEN the (supernumerary) blowers(s) must be switched off:

- Select the longest running blower(s) and stop it (them).
- Control the other blower(s) on the requested capacity.

IF

- No aeration is required

THEN

- Stop running blower(s).

Blower selection

Selection of the duty and standby blower takes place based on a duty rotation program in the Process Automation.

4.2.2 Settings

- Settings for blowers and control
- Insufficient air capacity settings
- Blower capacity scaling settings

4.2.3 Interlocks / alarms

- Insufficient air capacity (for each reactor) (available blower capacity lower than demand)
- No blower available for Reactor # (for each reactor) (*alarm*)

4.2.4 Presentation

- Blowers in current configuration for each header

4.3 Specification sheets

4.3.1 Blower V4601, V4602, V4603 and V4604

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- Capacity setpoint (Manual)
- Controller settings (gain, integration time)

Automation

Automatic

In automatic mode, the blowers are controlled as described in section 4.2.

Manual

In manual mode, the operator can start and stop the blower. The blower must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- thermal overload (TS)
- electric circuit failure
- safety switch off
- frequency inverter failure

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure
- Actual blower capacity

4.3.2 Blower selection valves AV46001, AV46002, AV46003

Settings

SCADA

- Manual-0-Auto
- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under section 4.2.

Manual

In manual control the operator can issue the following commands:

- open
- close

Interlocks

- check on feedback failure (valve opening / closing)
- electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

5 NEREDA® SLUDGE BUFFER

5.1 Introduction

The sludge of the Nereda® sludge buffer is pumped to the sludge thickener. The supernatant water is discharged to the influent buffer tank.

For the sludge buffer, 3 frequency controlled pumps are installed for sludge discharge and supernatant water discharge. For sludge discharge, the pump configuration is 1 duty + 2 standby, for supernatant discharge 2 duty + 1 standby.

In this chapter the phases, valves, instruments and equipment of sludge buffer 1 are described. The phases for sludge buffer 2 must be realized in the same way.

5.2 Automation

The sludge buffer is controlled by the Nereda® Controller, see description in section 3.2. If the communication between the Nereda® controller and the PLC should fail, it switches to an adjustable emergency recipe that is programmed into the PLC. See section 7 **Error! Reference source not found.** and the description in the document "Control Philosophy" (Appendix 4.1).

The pumps must be stopped if the level in the buffer is lower than an adjustable low level 49LT0001.

For the sludge buffer following phases are described:

- feed
- wait (settle)
- sludge discharge
- water discharge

Flow measurement failure

In case of flow meter failure, corrective actions must be taken in the Process Automation. The Nereda® Controller will still issue a flow setpoint to the Process Automation. In the Process Automation, the flow setpoint must be used to determine the number of pumps to be operated and the pump speed by using the theoretical pump capacity (Q-H curve table in Process Automation).

Sludge buffer not available

In case a sludge buffer is not available to receive sludge from the Nereda reactors, the Nereda® Controller generates a warning which will be displayed on the SCADA.

Interlocking

During the "RUN" states of the water and sludge discharge phases, the discharge must be paused in either of the following situations:

- Low level 49LT0001 sludge buffer.
- LowLow level 49LS0005 sludge buffer

If the measured values are above these switching points, then the discharge must be started again after an adjustable waiting time.

5.2.1 Phase feed

In this phase, the sludge buffer is filled by means of the sludge discharge phase of the Nereda® reactor, described in section 3.2.6. This is only the description for the start phase.

Parameters

- None

Start logic

- None

RUN logic

- None

Stop logic

- None

Operating conditions

1. Reserved, do not use.
2. No High level in the sludge buffer 49LT0001
3. HighHigh level switch 49LS0002 not activated
4. Level measurement 49LT0001 available
5. Spare
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

5.2.2 Phase wait

Parameters

- None

Start logic

- None

RUN logic

- None

Stop logic

- None

Operating conditions

1. Spare
2. Spare
3. Spare
4. Spare
5. Spare
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

5.2.3 Phase sludge discharge

The sludge discharge phase controls the sludge / supernatant discharge pumps P4901, P4902 and P4903. The discharge pumps must be PI-controlled on the flow setpoint from the Nereda® Controller.

Parameters

- Flow setpoint 'sludge discharge' [m³/h]

Start logic

- Open valve AV49004 AND/OR AV49005
- Start (duty) pump

RUN logic

- The (duty) pump will be in operation continuously on the flow setpoint from the Nereda® Controller.

Stop logic

- Stop pumps
- Close valve AV49004 AND AV49005

Operating conditions

1. Valve AV49004 OR AV49005 is available
2. Pump P4901 or P4902 or P4903 is available
3. Valve AV49002 is closed
4. *Reserved, do not use*
5. Valve AV49003 is closed
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

5.2.4 Phase water discharge

The water discharge phase controls the sludge / supernatant discharge pumps P4901, P4902 and P4903. The discharge pumps must be PI-controlled on the flow setpoint from the Nereda® Controller.

Parameters

- Flow setpoint 'supernatant water discharge' [m³/h]

Start logic

- Open valve AV49002
- Open valve AV49003
- Start pumps

RUN logic

- The pumps will be in operation continuously on the flow setpoint from the Nereda® Controller.

Stop logic

- Stop pumps
- Close valve AV49003
- Close valve AV49002

Operating conditions

1. Valve AV49002 is available
2. *Reserved, do not use*
3. Pump P4901 or P4902 or P4903 is available
4. Valve AV49004 AND AV49005 are closed
5. No High level in the influent buffer 40LT0001
6. No HH-level in the influent buffer 40LS0002
7. Level measurement 40LT0001 available
8. Valve AV49005 is available
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare
14. Spare
15. Spare
16. Spare

5.2.5 Settings

Adjustable values, as mentioned in the automation section above, must be made adjustable on SCADA.

5.2.6 Alarms

The status "Aborted" of each phase must be configured as an alarm

5.2.7 Presentation

Presentation via the web pages of the Nereda® Controller and as indicated in the document "Control Philosophy" (appendix 4.1).

5.3 Specification sheets

5.3.1 Sludge / supernatant discharge pumps P4901, P4902 and P4903

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- Capacity setpoint (Manual)
- Controller settings (gain, integration time)
- settings Low level for dry run protection

Automation

Automatic

In automatic mode, the pumps are controlled as described in section 5.2.

Manual

In manual mode, the operator can start and stop the pump. The pump must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- Thermal overload motor
- Electric circuit failure
- Safety switch off
- Frequency inverter failure
- LL-flow (time delayed)

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure
- Actual pump capacity

5.3.2 Valves Nereda® sludge buffer AV49002, AV49003, AV49004

Settings

SCADA

- Manual-0-Auto
- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under section 5.2.

Manual

In manual control the operator can issue the following commands:

- Open
- Close

Interlocks

- Check on feedback failure (valve opening / closing)
- Electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

5.3.3 Level measurement sludge buffer 49LT0001

SCADA

- Level [m]

switch points	action	Ad	Pv	Al	alarm text
HH	Abort / interlock Feed phase sludge buffer -> operation conditions not met	x		x	Level too High sludge buffer
LL	Stop sludge / supernatant discharge pumps	x		x	Level too Low sludge buffer
Failure measurement	Abort / interlock Feed phase sludge buffer -> operation conditions not met			x	Failure level measurement influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

5.3.4 Level switch sludge buffer 49LS0002

SCADA

- Level switch point

switch points	action	Ad	Pv	Al	alarm text
HH	Abort / interlock Feed phase sludge buffer -> operation conditions not met			x	Level too High sludge buffer
Failure measurement				x	Failure level switch sludge buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

5.3.5 Dry solids measurement discharged sludge 49QT0003

SCADA

- Dry solids [g/l]

switch points	Action	Ad	Pv	Al	alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure dry solids measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

5.3.6 Flow measurement discharged sludge 49FT0004

SCADA

- Flow [m³/h]

switch points	Action	Ad	Pv	Al	alarm text
HH		x		x	Flow too high
L		x		x	Flow low
LL	Stop sludge / supernatant discharge pumps	x		x	Flow too low (time delayed)
Deviation alarm		x		x	Deviation alarm discharge flow
Failure measurement	determine the number of pumps and pump speed by using the theoretical pump capacity, see 5.2			x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6 EFFLUENT

6.1 Introduction

The effluent from the Nereda® reactors is collected in the effluent gutter and drained to the effluent discharge.

6.2 Automation

- None

6.2.1 Settings

- See specification sheets 6.3

6.2.2 Alarms

- See specification sheets 6.3

6.2.3 Presentation

- See specification sheets 6.3

6.3 Specification sheets

6.3.1 Turbidity/dry solids measurement effluent 40QT2101

SCADA

- Dry solids [mg/l]

switch points	Action	Ad	Pv	Al	alarm text
H		x		x	Dry solids too High effluent
Failure measurement				x	Failure dry solids measurement effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

7 Fallback Control

7.1 Introduction

As described in the control philosophy, the fallback control is a safeguard located in and controlled by the Process Automation that is to be used in exceptional cases. This comprises of a series of equipment phases which are carried out sequentially on phase time basis.

The Fallback control consists of 2 parts:

1. Scheduling of the reactors. This determines which reactor is started with which phase in the case of transition to fallback control and during fallback control.
2. Execution of fallback recipe on a single reactor or sludge buffer.

Configuration

The commands that are sent to the phases by the emergency recipe have to be handled in the same way as the commands given from the Nereda® Controller.

7.2 Automation

The commands that are sent to the phases by Fallback recipe must be handled in the same way as the commands given from the Nereda® Controller.

In the Fallback recipe only one phase can be active at a time. The next phase may only be started when the previous one is completed (READY / INTERLOCKED/ ABORTED).

7.2.1 Fallback scheduler reactors

A Fallback Recipe for a reactor must be started when:

- The operation mode for the Nereda® Controller is switched via the process automation to 'Fallback' (automatically in case of communication failure or manually by operator)
- AND the Fallback Recipe for the reactor is set to 'On' (only possible if no phase Aborted)

When the operation mode of the Nereda controller is switched between different process control modes (see also A-1 Automation - Design Principles), all phases must first receive a STOP command except the 'feed' phase, which must remain active.

When the Fallback recipe is started, all the reactors in the Fallback control except for the reactors with an active 'feed' phase must complete an *init* step in which the influent grid is vented. The *init* step must only be completed only one time when the Fallback recipe is started.

Calculating the order of reactors

To calculate the order of reactors in each line the Process Automation must keep track of the last feed phase of each reactor of each line (MinutesSinceStartFeed) variable, this variable is reset when the Feed phase is starting. Based on this variable it will determine a reactor number for the first, second, third reactor*. This will also be determined with more reactors. The current feeding reactor should continue feeding, after this the reactor which fed the longest time ago must start.

*) Note: a reactor should be available to determine the order.

The scheduling of the reactors is based on the *StartFeedInterval*. Every *StartFeedInterval* a new reactor starts feeding. The *StartFeedInterval* can be calculated using the following formula:

$$StartFeedInterval = \frac{CT_{Nominal}}{N_{total\ reactors}}$$

Discontinuous feed

In discontinuous feed, the lines also have an order, so the lines should always feed in the same order. During determining the order of reactors, the reactors should be added to the list of reactors in the same line order. As a result, reactors in a line with less reactors available can have a shorter cycle time.

Example: Single line discontinuous feed

In this example there is a single line present with three reactors, each with the following *MinutesSinceStartFeed*:

- Reactor 1: MinutesSinceStartFeed = 35 min, is not feeding
- Reactor 2: MinutesSinceStartFeed = 140 min, is not feeding
- Reactor 3: MinutesSinceStartFeed = 216 min, is not feeding

With the times above the order of reactors to start feeding would be the following: [3, 2, 1]. Reactor 3 has the highest *MinutesSinceStartFeed*, and is therefore the first reactor in order.

7.2.2 Fallback control Nereda® reactor

The Fallback control for the reactor is filled in on the SCADA/ HMI, the table below shows what should be available to fill in.

Table 5 - Fallback control Nereda® Reactor, discontinuous feed

Step	Active phase	Duration / capacity (adjustable via SCADA)
0	Idle	
1	Feed	0....60 minutes Flow setpoint: 0....1200 m ³ /h (OR stop on InfluentBuffer level <= FB_IBL1)
2	Vent sludge grid	0....15 minutes
3	Sludge discharge**	0....30 minutes OR stop on Reactor level <= FB_RL1 Skipped if no SB available
4	Lower level	0....15 minutes OR stop on Reactor level <= FB_RL1
5	Aerate*	0....10 minutes air fill time airlock sludge decant 0....100% aeration capacity
6	Vent aeration grid	0....15 minutes
7	Wait	0....60 minutes
8	Vent sludge grid	0....15 minutes
9	Sludge discharge**	0....30 minutes OR stop on Reactor level <= FB_RL2 Skipped if no SB available or discharge is already done in step 3
10	Wait	0....60 minutes
I	Pre-set cycle time for all reactors (CTnominal):: 30....360 minutes	
II	FB_RL1: Level on which the LowerLevel (Step 4) and SludgeDischarge (Step 3) phase will stop	
III	FB_RL2: Level on which the SludgeDischarge (Step 9) will stop	
IV	FB_IBL1: Level in influent buffer on which the feed (Step 1) will stop	
**)	If the recipe has had a sludge discharge in step 3, it should skip the sludge discharge in step 9. Depending on the choice, 0 min for the phase time must be entered in step 3 or in step 9.	

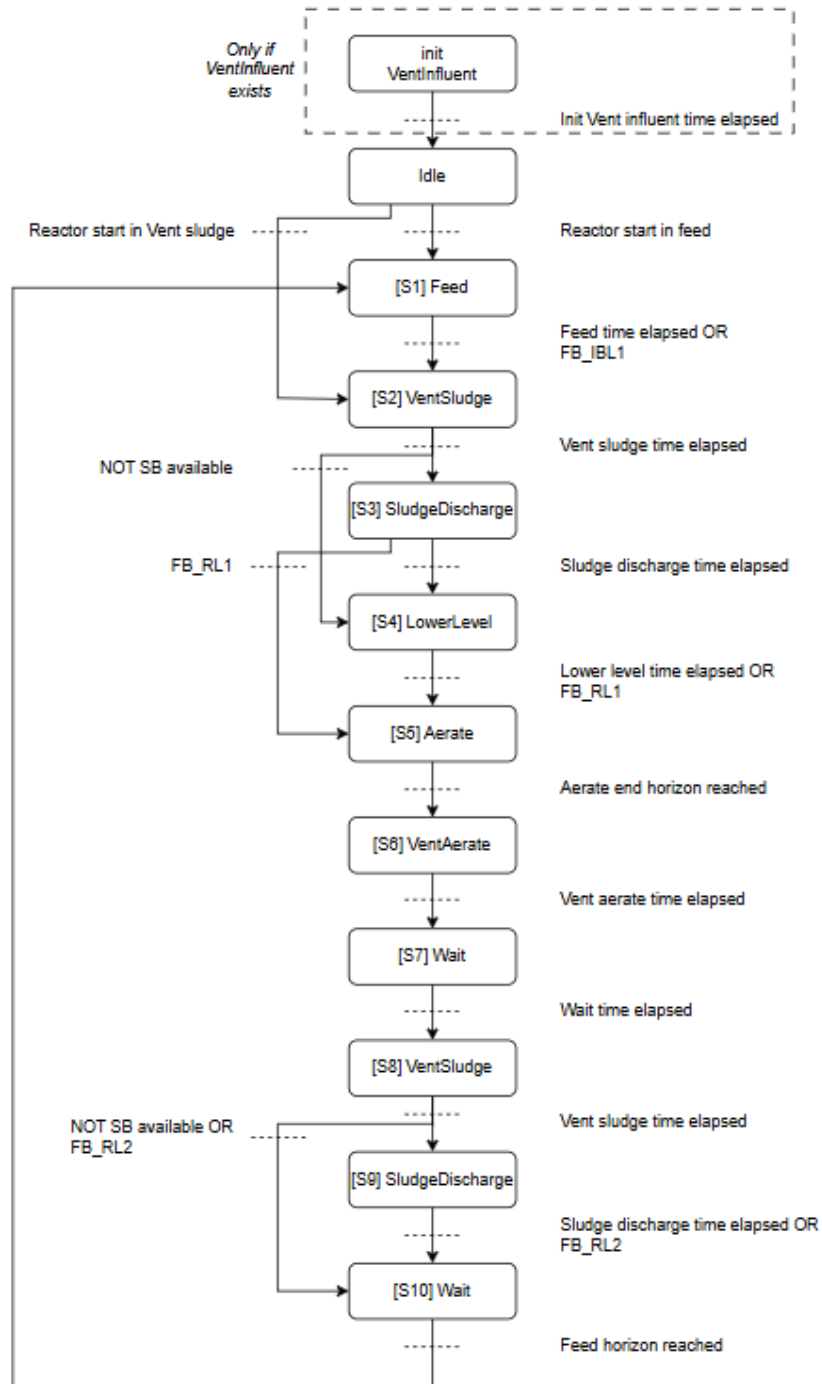


Figure 2 - Flowdiagram Reactor Discontinuous feed

Remaining feed time

After starting the fallback control and the calculation of order of reactors, one of the reactors might continue feeding. In this case the reactor should be feeding for $\text{RecipeFeedTime} - \text{TimeFedBeforeFallback}$. If this time becomes less than zero the reactor stops feeding.

Table 6 - Discontinuous feed remaining feed time.

Reactor in Order	Proces Values	
	<i>RecipeFeedTime</i> [60 min]	<i>TimeFedBeforeFallback</i> [40 min]
1	(60) – 40 → 20 min	

Allow settle before feeding when switching to fallback control

Due to the nature of discontinuous feed, it might occur that a reactor is aerating in normal operation and is the first reactor in order when switching to fallback control. To prevent a reactor from feeding right after aerating, the feed phase is delayed. The Process Automation must keep track of time since stopping of the last aeration phase of each reactor.

Using the following formula, a settle time can be calculated:

$$SettleTime = MinimumSettleTime - TimeSinceLastAerate$$

FeedHorizon single line

During discontinuous flow it might happen no reactor is feeding at all, this is calculated using the *FeedInterval* which is calculated using:

$$CT_{nominal} / N_{total\ reactors}$$

When the fallback recipe starts the current feeding reactor keeps feeding for its remaining feed time. When no reactor is feeding the *NextFeedHorizon* is calculated as followed.

When starting the recipe the *RecipeElapsedTime* must be saved as well, this time starts counting from the moment that the first reactor in order starts feeding. The time is kept for each reactor and reset when the reactor in question starts feeding again. The calculation returns a decreasing time until step 1 must start again, for the first reactor in order this value will be 0 (see examples below). The formulas change based on the times the reactors have been fed in fallback recipe. Once a feed phase is started, the value *TimesFed* is incremented by 1.

Calculate *NextFeedHorizon* on *TimesFed* = 0:

$$NextFeedHorizon = (i - 1) * StartFeedInterval - FallbackElapsedTime + SettleTime$$

NextFeedHorizon is rounded to zero if it gets below zero

If there is a reactor feeding when switching over to the fallback control, the remaining time is calculated using the formula below. If the *RemainingFeedTime* ≤ 0 then the reactor stops feeding.

$$RemainingFeedTime = RecipeFeedTime - TimeFedBeforeFallback$$

where *i* is the reactor in order.

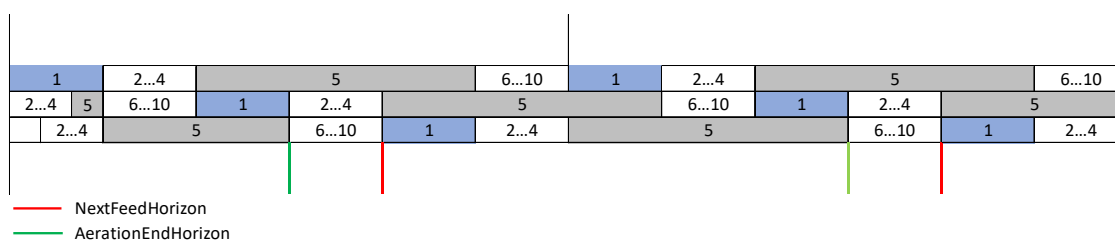
Table 7 - Discontinuous flow wait time before start feed

Reactor in Order	Proces Values		
	<i>StartFeedInterval</i> [90 min]	<i>ElapsedFallbackTime</i> [0 min]	<i>SettleTime</i> [10 min]
1	((1-1) * 90) + 10 → 10 min		
2	((2-1) * 90) + 10 → 100 min		
3	((3-1) * 90) + 10 → 190 min		

After a first cycle of feeding the following formula is used to calculate the *NextFeedHorizon*.

Calculate *NextFeedHorizon* on *TimesFed* > 0:

$$NextFeedHorizon = (N_{AvailableReactors} * StartFeedInterval) - RecipeElapsedTime$$

**Figure 3 - Fallback control scheduling Discontinuous feed**

Feeding multiple lines

When feeding multiple lines during discontinuous feed there are multiple calculations that need to be done. As stated in the beginning the *StartFeedInterval* is calculated using the formula:

$$StartFeedInterval = \frac{CT_{Nominal}}{N_{total\ reactors}}$$

This value gives the time until a reactor starts feeding, to calculate the time between lines to feed the following formula is used:

$$StartFeedIntervalLine = \frac{CT_{Nominal}}{N_{total\ reactors}} * NrOfLines$$

Using this *StartFeedIntervalLine* it is possible to calculate a feed horizon for each line, for this the following formula is used:

$$NextFeedHorizonLine = (i - 1) * StartFeedIntervalLine - ElapsedFallbackTime$$

Where *i* is the line in order

NextFeedHorizonLine is rounded to zero if it gets below zero.

If there is a reactor feeding when switching over to the fallback control, the remaining time is calculated using the formula below. If the *RemainingFeedTime* ≤ 0 then the reactor stops feeding.

$$RemainingFeedTime = RecipeFeedTime - TimeFedBeforeFallback$$

When the *NextFeedHorizonLine* is calculated, this can be used to calculate the *NextFeedHorizon* for each reactor. This can be calculated using the following formula:

$$NextFeedHorizon = ((i - 1) * CT_{Nominal}) + NextFeedHorizonLine$$

Filling in time before feeding

For each reactor that is waiting for feeding, there is a check in place if the time could be used to aerate. The check can be done by using the following formula:

$$AerateBeforeFeeding [0/1] = (NextFeedHorizon - T3 - T4 - T6 - T7 - T8 - T9 - T10) \\ \geq MinimumAerateTime$$

The reactor that is starting the soonest and is available for aerating before feeding, starts in step 2 of the fallback sequence. The following reactor that is available to do this starts in step 2 after the previous reactor is done.

Aeration during the fallback recipe

The main goal of the aeration time is to fill up the remaining time before starting the second waiting phase in a cycle. In order to do this, it is needed to calculate a horizon in which the aeration phase has to stop. This horizon can be calculated using the following formula, in which *RecipeElapsedTime* is the time elapsed since starting the Feed phase. When *AerationEndHorizon* = 0, the Aeration phase should stop and switch to the next phase.

$$AerationEndHorizon = NextFeedHorizon - (T_6 + T_7 + T_8 + T_9 + T_{10})$$

Exceptions during the process

During the process there is a possibility something goes wrong, depending on what a reactor is doing, it should abort and be taken out of fallback recipe*. A reactor aborts when:

When a phase aborts except for the SludgeDischarge and LowerLevel phase

When a phase should start but is interlocked except for the SludgeDischarge and LowerLevel phase

*) Note: when this happens, all calculations should be performed again.

7.2.3 Fallback control Sludge buffer

The fallback control for the sludge buffer must be activated simultaneously with the sludge discharge phase of the Nereda® reactor.

The fallback control for the Sludge buffer can consist of the following steps:

Table 8 - Fallback control Nereda® Sludge buffer

Step	Active phase	Duration / capacity (adjustable via SCADA)
0	Idle	Start in Wait phase if the SludgeBuffer level $\geq$ RestartLevel
1	Feed	0...30 minutes
2	Wait	0...30 minutes
3	Sludge discharge	0...30 minutes 0...80 m ³ /h OR stop on Sludge buffer level $\leq$ FB_SBL1
4	Supernatant discharge	0...30 minutes 0...320 m ³ /h OR stop on Sludge buffer level $\leq$ FB_SBL2
I	FB_SBL1: Level on which the sludge discharge will stop (step 3)	
II	FB_SBL2: Level on which the supernatant discharge will stop (step 4)	

If an active phase reaches the Aborted state, the fallback control for this sludge buffer must be aborted.

When the feed phase aborts/ stops the sludge discharge phase of the connected reactor should perform a normal stop of the phase. When the sludge discharge phase of the reactor aborts or stops, the Sludge Buffer fallback control should start step 2.

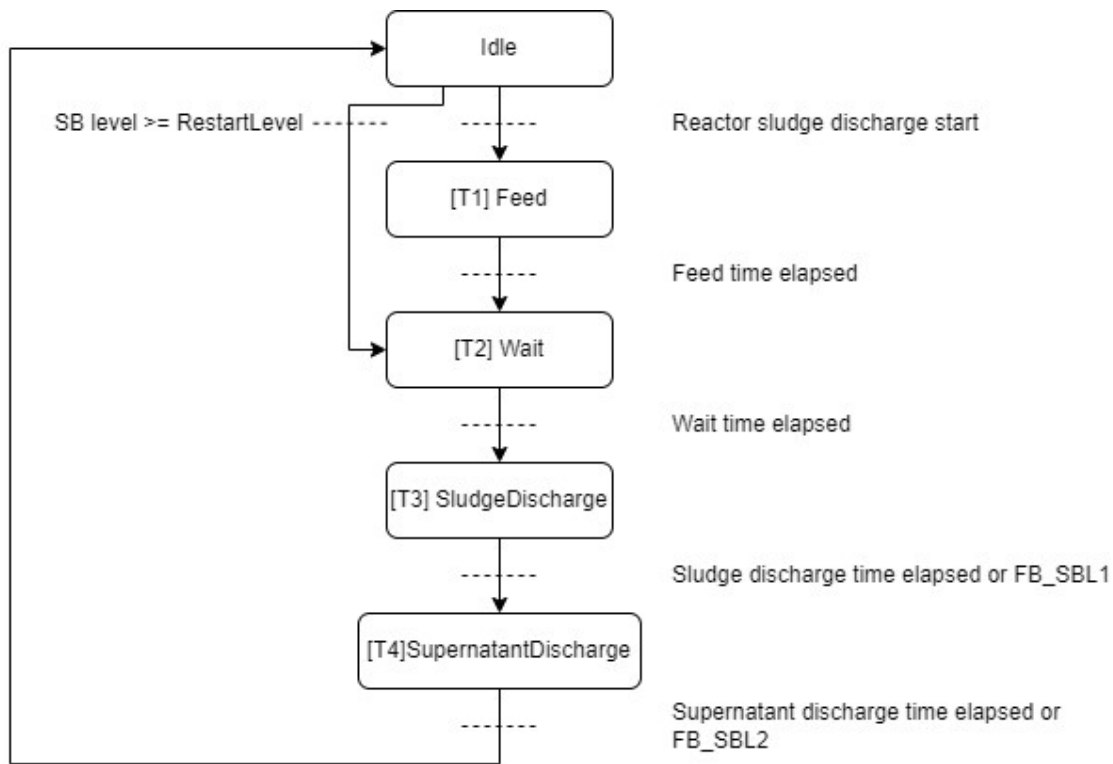


Figure 4 - Fallback control scheduling

7.2.4 Settings

On SCADA the operator must be able to adjust the following parameters for the Fallback control:

- Nereda® Fallback control start-up delay time
- Nereda® reactor cycle time
- Reset button for each reactor/ sludge buffer
- NrOfRequiredFeedingReactors
- For the Nereda® reactors and Nereda® sludge buffers:
- See table in sections [7.2.2/ Error! Reference source not found.](#) and 7.2.3

7.2.5 Presentation

Reactor

At least the following objects must be visualized on the SCADA:

- Input as shown in table from paragraph 7.2.2/ **Error! Reference source not found.**
- NextFeedHorizon [minutes]
- AerationEndHorizon [minutes]
- ReactorOrder [1, 2, 3,...]

- RecipeElapsedTime [minutes]
- Actual Air capacity [%]
- Actual Feed Flow [m³/h]¹
- MinimumSettleTime [minutes]
- TimeSinceLastAerate [minutes]

Sludge Buffer

At least the following objects must be visualized on the SCADA:

Input as shown in table from paragraph **Error! Reference source not found.**

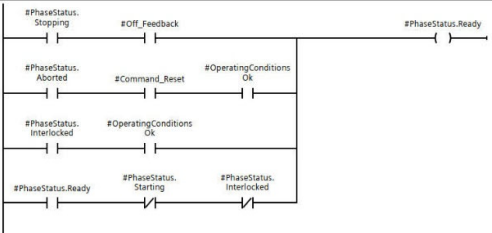
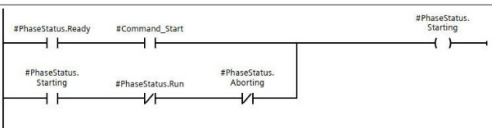
RecipeElapsedTime [minutes]

Actual SludgeDischargeFlow [m³/h]

Actual WaterDischargeFlow [m³/h]

¹ Only with Discontinuous flow

Appendix A Programming example Equipment Phase state diagram in Ladder & Structured Text

Totally Integrated Automation Portal					
AquasuiteControlPhase_LAD [FB3]					
AquasuiteControlPhase_LAD Properties					
General					
Name	AquasuiteControl-Phase_LAD	Number	3	Type	FB
Language	LAD				
Numbering Information					
Title	Aquasuite Control Phase	Author		Comment	
Version	0.1	User-defined ID		Family	
Variables					
Name	Data type	Default value	Retain	Accessible from HMI	Visible in HMI
Input					
OperatingConditionsOk	Bool	false	Non-retain	False	False
On_Feedback	Bool	false	Non-retain	False	False
Off_Feedback	Bool	false	Non-retain	False	False
Command_Start	Bool	false	Non-retain	False	False
Command_Stop	Bool	false	Non-retain	False	False
Command_Reset	Bool	false	Non-retain	False	False
Output					
Ready	Bool	false	Non-retain	False	False
Starting	Bool	false	Non-retain	False	False
Run	Bool	false	Non-retain	False	False
Stopping	Bool	false	Non-retain	False	False
Aborting	Bool	false	Non-retain	False	False
Aborted	Bool	false	Non-retain	False	False
Interlocked	Bool	false	Non-retain	False	False
InOut					
Static					
PhaseStatus					
Ready	Bool	false	Retain	False	False
Starting	Bool	false	Retain	False	False
Run	Bool	false	Retain	False	False
Stopping	Bool	false	Retain	False	False
Aborting	Bool	false	Retain	False	False
Aborted	Bool	false	Retain	False	False
Interlocked	Bool	false	Retain	False	False
Temp					
Constant					
Network 1: Ready					
					
Symbol	Address	Type	Comment		
#Command_Reset		Bool			
#Off_Feedback		Bool			
#OperatingConditions.Ok		Bool			
#PhaseStatus.Aborted		Bool			
#PhaseStatus.Interlocked		Bool			
#PhaseStatus.Ready		Bool			
#PhaseStatus.Starting		Bool			
#PhaseStatus.Stopping		Bool			
Network 2: Starting					
					

Totally Integrated Automation Portal			
Symbol	Address	Type	Comment
#Command_Start		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	

Network 3: Run

Symbol	Address	Type	Comment
#On_Feedback		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

Network 4: Stopping

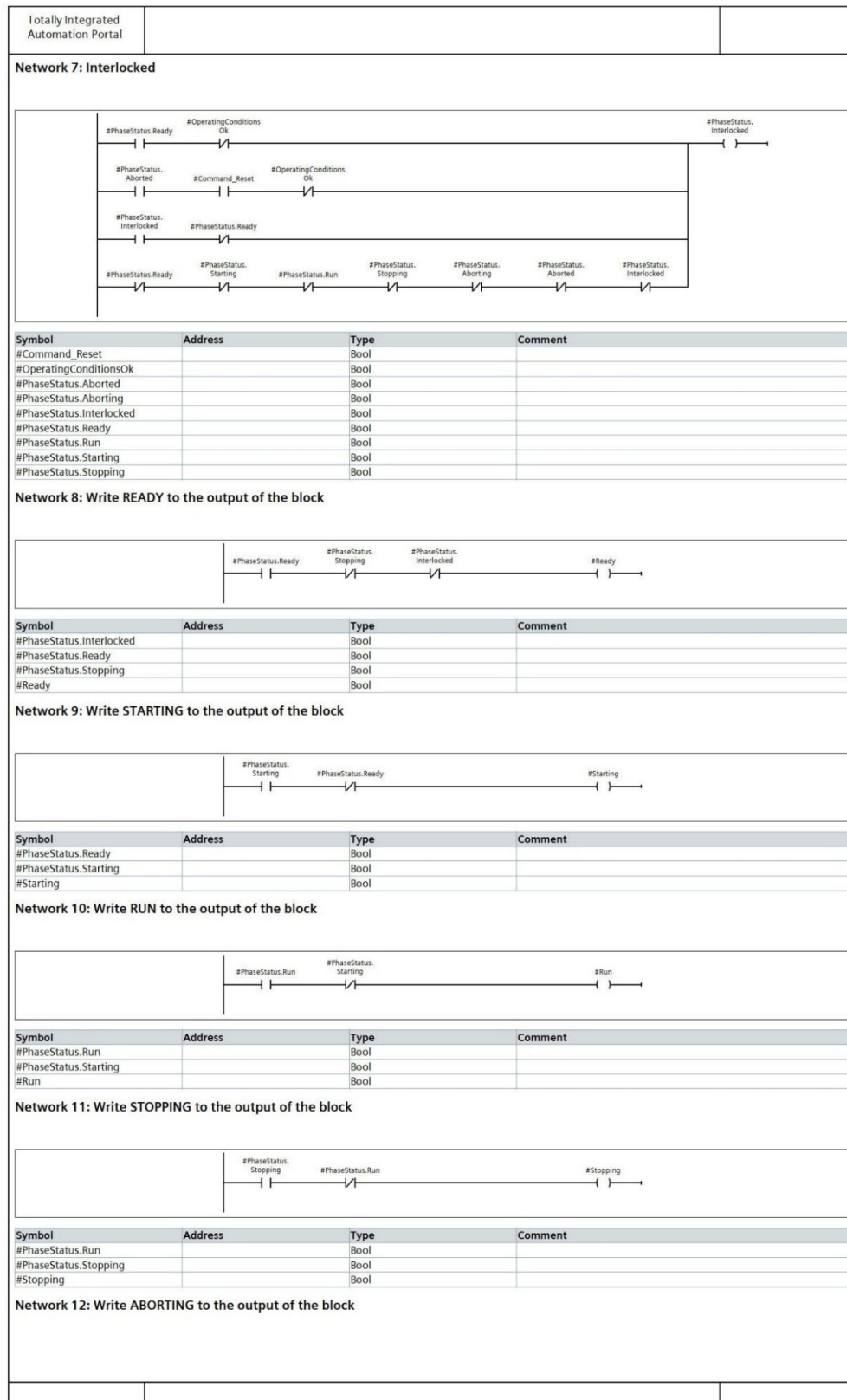
Symbol	Address	Type	Comment
#Command_Stop		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Stopping		Bool	

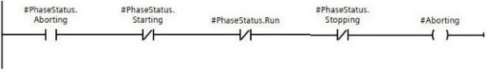
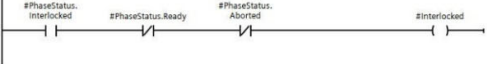
Network 5: Aborting

Symbol	Address	Type	Comment
#OperatingConditionsOk		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

Network 6: Aborted

Symbol	Address	Type	Comment
#Off_Feedback		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Interlocked		Bool	
#PhaseStatus.Ready		Bool	



Totally Integrated Automation Portal																									
																									
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 25%;">Symbol</th> <th style="width: 25%;">Address</th> <th style="width: 25%;">Type</th> <th style="width: 25%;">Comment</th> </tr> </thead> <tbody> <tr><td>#Aborting</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Aborting</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Run</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Starting</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Stopping</td><td></td><td>Bool</td><td></td></tr> </tbody> </table>		Symbol	Address	Type	Comment	#Aborting		Bool		#PhaseStatus.Aborting		Bool		#PhaseStatus.Run		Bool		#PhaseStatus.Starting		Bool		#PhaseStatus.Stopping		Bool	
Symbol	Address	Type	Comment																						
#Aborting		Bool																							
#PhaseStatus.Aborting		Bool																							
#PhaseStatus.Run		Bool																							
#PhaseStatus.Starting		Bool																							
#PhaseStatus.Stopping		Bool																							
Network 13: Write ABORTED to the output of the block																									
																									
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 25%;">Symbol</th> <th style="width: 25%;">Address</th> <th style="width: 25%;">Type</th> <th style="width: 25%;">Comment</th> </tr> </thead> <tbody> <tr><td>#Aborted</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Aborted</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Aborting</td><td></td><td>Bool</td><td></td></tr> </tbody> </table>		Symbol	Address	Type	Comment	#Aborted		Bool		#PhaseStatus.Aborted		Bool		#PhaseStatus.Aborting		Bool									
Symbol	Address	Type	Comment																						
#Aborted		Bool																							
#PhaseStatus.Aborted		Bool																							
#PhaseStatus.Aborting		Bool																							
Network 14: Write INTERLOCKED to the output of the block																									
																									
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 25%;">Symbol</th> <th style="width: 25%;">Address</th> <th style="width: 25%;">Type</th> <th style="width: 25%;">Comment</th> </tr> </thead> <tbody> <tr><td>#Interlocked</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Aborted</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Interlocked</td><td></td><td>Bool</td><td></td></tr> <tr><td>#PhaseStatus.Ready</td><td></td><td>Bool</td><td></td></tr> </tbody> </table>		Symbol	Address	Type	Comment	#Interlocked		Bool		#PhaseStatus.Aborted		Bool		#PhaseStatus.Interlocked		Bool		#PhaseStatus.Ready		Bool					
Symbol	Address	Type	Comment																						
#Interlocked		Bool																							
#PhaseStatus.Aborted		Bool																							
#PhaseStatus.Interlocked		Bool																							
#PhaseStatus.Ready		Bool																							

Totally Integrated Automation Portal							
AquasuiteControlPhase_ST [FB2]							
AquasuiteControlPhase_ST Properties							
General							
Name	/AquasuiteControlPhase_ST	Number	2	Type	FB		
Numbering	automatic	Language	SCL				
Information							
Title		Author		Comment	Family		
Version	0.1	User-defined ID					
Name	Data type	Default value	Retain	Accessible from HMI	Visible in HMI	Setpoint	Comment
▼ Input							
OperatingConditionsOK	Bool	false	Non-retain	False	False	False	
On_Feedback	Bool	false	Non-retain	False	False	False	
Off_Feedback	Bool	false	Non-retain	False	False	False	
Command_Start	Bool	false	Non-retain	False	False	False	
Command_Stop	Bool	false	Non-retain	False	False	False	
Command_Reset	Bool	false	Non-retain	False	False	False	
▼ Output							
Status	Int	0	Non-retain	False	False	False	
Ready	Bool	false	Non-retain	False	False	False	
Starting	Bool	false	Non-retain	False	False	False	
Run	Bool	false	Non-retain	False	False	False	
Stopping	Bool	false	Non-retain	False	False	False	
Aborting	Bool	false	Non-retain	False	False	False	
Aborted	Bool	false	Non-retain	False	False	False	
Interlocked	Bool	false	Non-retain	False	False	False	
InOut							
▼ Static							
PhaseStatus	Int	7	Retain	False	False	False	
Temp							
▼ Constant							
cReady	Int	0					
cStarting	Int	1					
cRun	Int	2					
cStopping	Int	3					
cAborting	Int	5					
cAborted	Int	6					
cInterlocked	Int	7					
<pre> 0001 #Ready := false; 0002 #Starting := false; 0003 #Run := false; 0004 #Stopping := false; 0005 #Aborting := false; 0006 #Aborted := false; 0007 #Interlocked := false; 0008 0009 CASE #PhaseStatus OF 0010 0: // Ready 0011 #Ready := true; 0012 0013 IF #Command_Start THEN 0014 #PhaseStatus := #cStarting; 0015 END_IF; 0016 0017 IF NOT #OperatingConditionsOK THEN 0018 #PhaseStatus := #cInterlocked; 0019 END_IF; 0020 1: // Starting 0021 #Starting := true; 0022 0023 IF #On_Feedback THEN 0024 #PhaseStatus := #cRun; 0025 END_IF; 0026 0027 IF NOT #OperatingConditionsOK THEN 0028 #PhaseStatus := #cAborting; 0029 END_IF; 0030 2: // Run 0031 #Run := true; 0032 0033 IF #Command_Stop THEN 0034 #PhaseStatus := #cStopping; 0035 END_IF; 0036 0037 IF NOT #OperatingConditionsOK THEN 0038 #PhaseStatus := #cAborting; 0039 END_IF; 0040 3: // Stopping 0041 #Stopping := true; 0042 0043 IF #Off_Feedback THEN </pre>							

Totally Integrated Automation Portal

```
0044     #PhaseStatus := #cReady;
0045     END_IF;
0046
0047     IF NOT #OperatingConditionsOK THEN
0048         #PhaseStatus := #cAborting;
0049     END_IF;
0050 5:    // Aborting
0051     #Aborting := true;
0052
0053     IF #Off_Feedback THEN
0054         #PhaseStatus := #cAborted;
0055     END_IF;
0056 6:    // Aborted
0057     #Aborted := true;
0058
0059     IF #Command_Reset AND #OperatingConditionsOK THEN
0060         #PhaseStatus := #cReady;
0061     END_IF;
0062
0063     IF #Command_Reset AND NOT #OperatingConditionsOK THEN
0064         #PhaseStatus := #cInterlocked;
0065     END_IF;
0066 7:    // Interlocked
0067     #Interlocked := true;
0068
0069     IF #OperatingConditionsOK THEN
0070         #PhaseStatus := #cReady;
0071     END_IF;
0072 ELSE
0073     #PhaseStatus := #cInterlocked;
0074 END_CASE;
0075
0076 #Status := #PhaseStatus;
```

Symbol	Address	Type	Comment
#Aborted		Bool	
#Aborting		Bool	
#cAborted	6	Int	
#cAborting	5	Int	
#cInterlocked	7	Int	
#Command_Reset		Bool	
#Command_Start		Bool	
#Command_Stop		Bool	
#cReady	0	Int	
#cRun	2	Int	
#cStarting	1	Int	
#cStopping	3	Int	
#Interlocked		Bool	
#Off_Feedback		Bool	
#On_Feedback		Bool	
#OperatingConditionsOK		Bool	
#PhaseStatus		Int	
#Ready		Bool	
#Run		Bool	
#Starting		Bool	
#Status		Int	
#Stopping		Bool	